भारी उद्योग मंत्रालय
MINISTRY OF
HEAVY INDUSTRIES
सत्यमेव जयते

SASTRA
ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act,1956)
THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

Electronic Manufacturing System & Product Compliance Training

Ki-CAD PCB Design Assignment

Name :MAHIJA. G

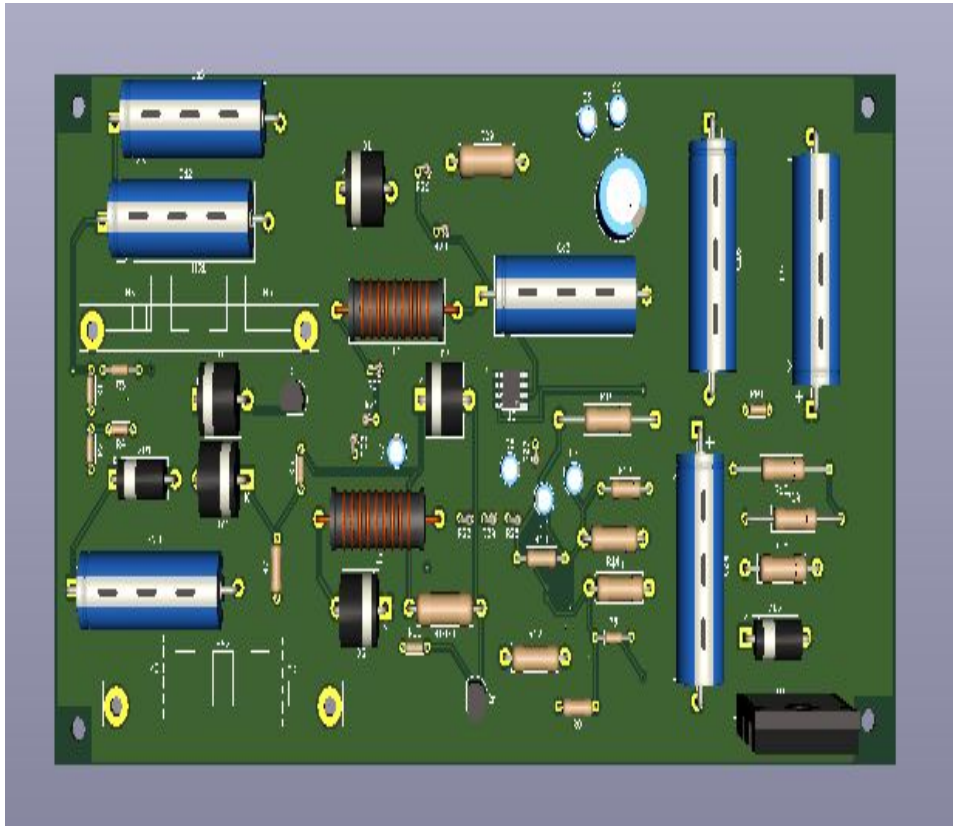
Branch :ECE

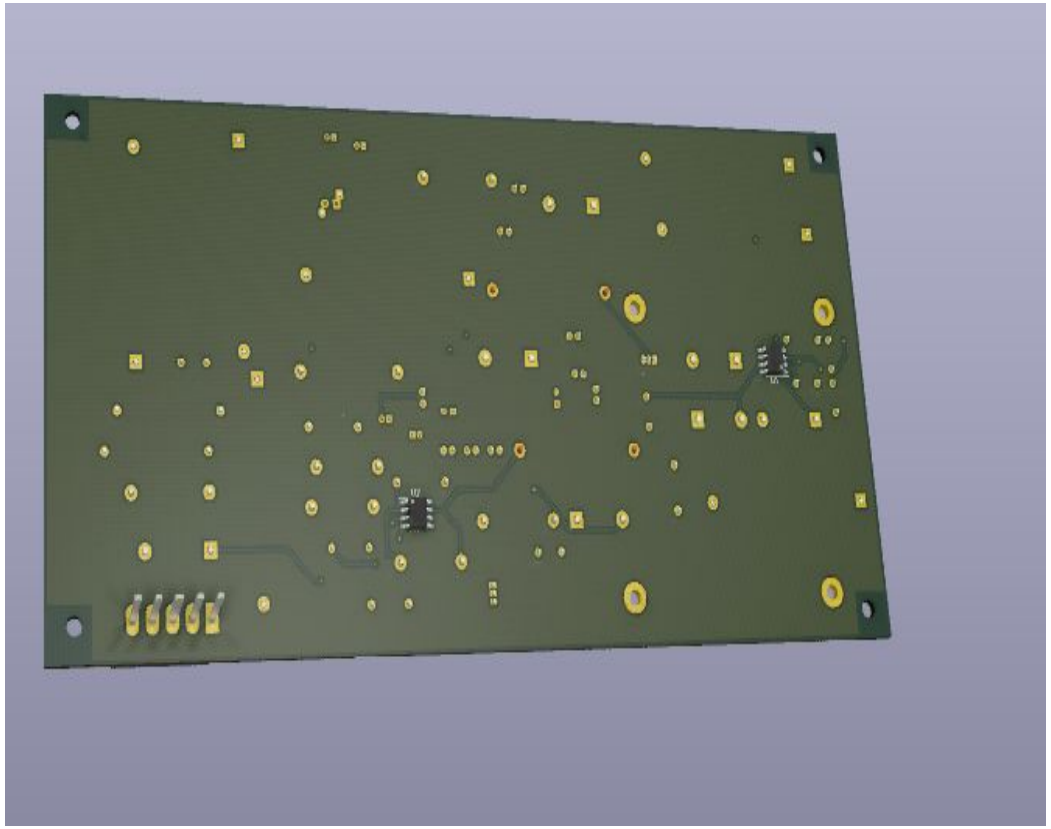
SEM/YEAR :4th-2nd

Reg . Number :12004149

Institution Name :SASTRA DEEMED TO BE UNIVERSITY

Valley switching boost power factor correction (PFC)



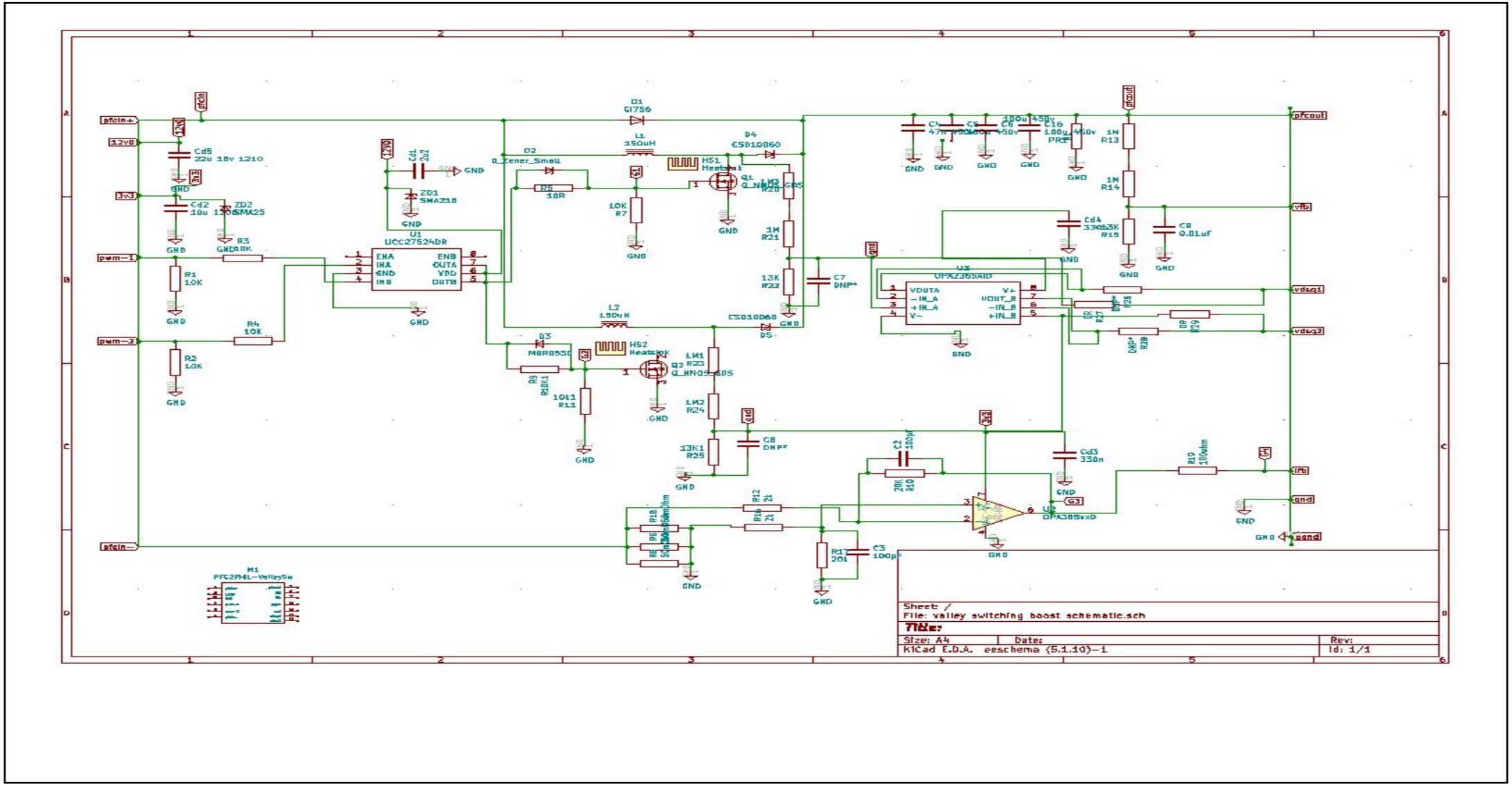


https://www.ti.com/tool/TIDM-1022

DESCRIPTION OF THE CIRCUIT

1. **Objective**: The design aims to enhance the performance of a Boost PFC converter, particularly under light load conditions where efficiency and Total Harmonic Distortion (THD) standards are challenging to meet.
2. **Controller**: Utilizes the integrated digital control feature (Type-4 PWM) of the **Piccolo™ F280049 microcontroller (MCU)**.
3. **Key Features**:
    - **Interleaved, 750 W, Two-Phase Boost PFC Stage**: Supports a wide input voltage range (95–260 Vrms, 47–63 Hz) with a programmable output voltage (380 V DC nominal).
    - **Switching Frequency**:
        - Normal Conditions (Load > 10%): **200 kHz**.
        - Light Load (< 10%): Variable Pulse Width Modulation (PWM) switching in the range of **150–330 kHz**.
    - **Control Techniques**:
        - **Phase Shedding**: Efficiently manages load variations.
        - **Valley Switching**: Optimizes performance during light loads.
        - **Valley Skipping**: Further enhances efficiency.
        - **Zero Voltage Switching (ZVS)**: Reduces switching losses.
4. **Applications**:
    - **Electric Vehicles (EVs) Charging**
    - **Servers**
    - **Telecom Rectifiers**
    - **Industrial Power Supplies**
5. **Software**: The provided software accelerates time to market.

SCHEMATIC


pfcin+
12v0
3v3
pwm-1
pwm-2
pfcin-
U1
UCC27524DR
D1
GI756
L1
150uH
D4
CSD10060
HS1
HeatSink
Q1
Q_NMOS_GDS
L2
150uH
D5
CSD10060
HS2
HeatSink
Q2
Q_NMOS_GDS
D3
MBR0530
U3
OPA2365AID
U5
OPA365xD
M1
PFC2PHL-ValleySW
pfcout
vfb
vdsq1
vdsq2
ifb
gnd
pgnd
Sheet /
File: valley switching boost schematic.sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. eeschema (5.1.10)-1
Id: 1/1

BOM

S.NO	COMPONENT DESCRIPTION	FOOTPRINT	COMPONENT DESIGNATION	MANUFACTURER PART NUMBER	ODERCODE	UNIT PRICE	REQUIRED QTY	COST FOR REQUIRED QUANRTITY
1	CAP CER 100PF 50V 5% NP0 0805	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm	C2,C3	CC0805JRNPO9BN101	3606028	3.23	2	6.43
2	CAP FILM 0.047UF 630VDC RADIAL	Capacitor_THT:CP_Radial_D10.0mm_P2.50mm_P5.00mm	C4	DMM6S47K-F		177.47	1	177.45
3	Elec. Cap - 100uf, 450V, Alum, radial	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm	C5,C,C16	EKXG51ELL101MM40S		361.57	3	
4	CAP CER 10nF 50V 10% X7R 0805	Capacitor_THT:CP_Axial_L29.0mm_D10.00mm_P35.00mm_Horizontal	C9	KGM21NR71H103KT		8.33	1	0.33

5	Cer. Cap - 2u2, 25V, X7R, 0805	Capacitor_THT:CP_Axial_L29.0mm_D10.00mm_P35.00mm_Horizontal	Cd1	GRM219R61E225KA12D	36.66	1	36.66
6	CAP CER 10UF 16V 10% X7R 1206	Capacitor_THT:CP_Axial_L29.0mm_D10.00mm_P35.00mm_Horizontal	Cd2	C1206C106K4RAC700	29.16	1	29.16
7	Cer. Cap- 330n, 16V, X5R, 0805	Capacitor_THT:CP_Axial_L29.0mm_D10.00mm_P35.00mm_Horizontal	Cd3	C2012X7R1H334K125AA	14.16	1	14.16
8	Cer. Cap - 22u, 16V, X5R, 1206	Capacitor_THT:CP_Axial_L29.0mm_D10.00mm_P35.00mm_Horizontal	Cd5	EMK316BJ226ML-T	41.66	1	41.66
9	Res. - 10K, 5%, 0.1W, 0805	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.0mm_Horizontal	R1,R2,RR17,R11	RC0805JR-0710KL	8.33	4	33.32
10	es. - 10R, 5%, 0.1W, 0805	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.0mm_Horizontal	R3,R4,R5, R9	RC0805JR-0710KL	8.33	4	33.32
11	Res 0.050 OHM 1W 1% 2512	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	R6,R8, R18	CRA2512-FZ-R050ELF	53.06	3	159.18
12	Res. - 20K, 1%, 1/8W, 0805	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	R10,R17	RC0805FR-0720KL	8.33	2	16.66

13	Res. - 1M, 1%, 0.1W,	Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P20.32mm_Horizontal	R13, R14	RC0805FR-071ML	8.33	2	1
14	Res. - 13.0K, 1%, 0.1W, 0805	Resistor_THT:R_Axial_DIN0411_L9.9mm_D3.6mm_P1.24mm_Horizontal	R15	RC0805FR-0713KL	8.33	1	
15	Res. - 2K, 1%, 0.1W, 0805	Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P20.32mm_Horizontal	R12, R14	RC0805FR-072KL	8.33	2	1
16	Res. - 100R, 5%, 0.1W, 0805	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P1.90mm_Vertical	R19				
17	DNP	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal	PR1				
18	DIODE ZENER 500MW 5.1V SOD123	Diode_THE:D_DO-201_P12.70mm_Horizontal	ZD2	NZH5V1B,115	21.66	1	2
19	Dual Power Mosfet Driver HS, 4A, 8-SOIC, UCC27524DR	Package_SO:SOIC-8_3.9*4.9mm_P1.27mm	U1,U3	UCC27524DR	105.8	2	211
20	Amplifier, OPA365, SOT-23-5	Package_SO:SOIC-8_3.9*4.9mm_P1.27mm	U2	OPA365AIDBVR	179.95	1	17
21	MOSFET N-CH 650V 12A TO220	Package_TO_SOT_THT:TO-92_Inline	Q1,Q2	IPP60R180P7XKSA1	215.31	2	43
22	RECTIFIER 6A 600V P600 AXIAL	Diode_THT:D_5KP_P10.16mm_Horizontal	D1	GI756-E3/54	74.98	1	7

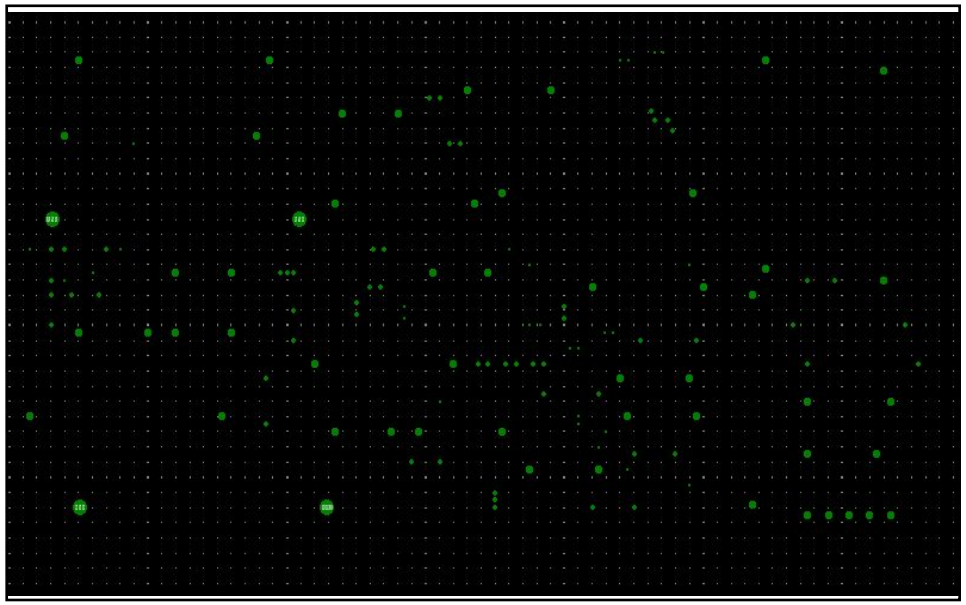
23	DIODE SCHOTTKY 30V 0.5A SOD-123	Diode_THT:D_5KP_P10.16mm_Horizontal	D2,D3	STPS0530Z	39.98	2	79.96
24	C3D10060G, DIODE SCHOTTKY 600V 10A TO263-2	Diode_THT:D_5KP_P10.16mm_Horizontal	D4,D5	C3D10060G	506.66	2	1013.32
25	Heat Sink, 2.0", Wakefield, Outline 34.92mm x 12.70mm, Height 2"	Heastsink:Heatsink_SheetType_50*7mm_2FIXATIONS	HS1,HS2	637-20ABPE	120.9	2	241.8
26	INDUCTOR CTX16-18702-R COOPER BUSSMAN		L1,L2	CTX16-18702-R	880.3	2	1760.6

Product Cost

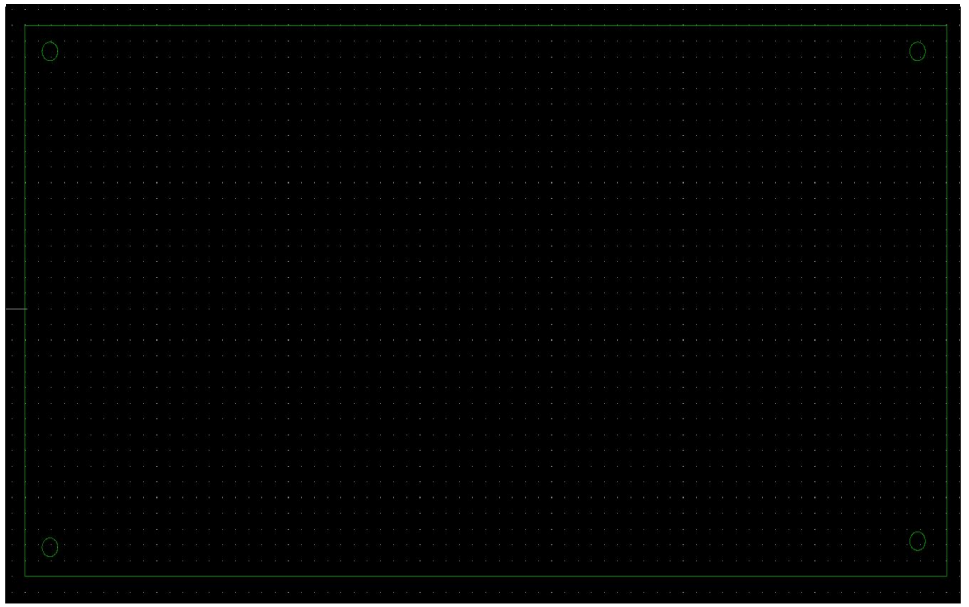
S.No	description	Cost (Rs)
1	total order value	13050.00
2	tax rate(%)	18
3	tax price	2349
	Total Cost	15399

GERBER FILES

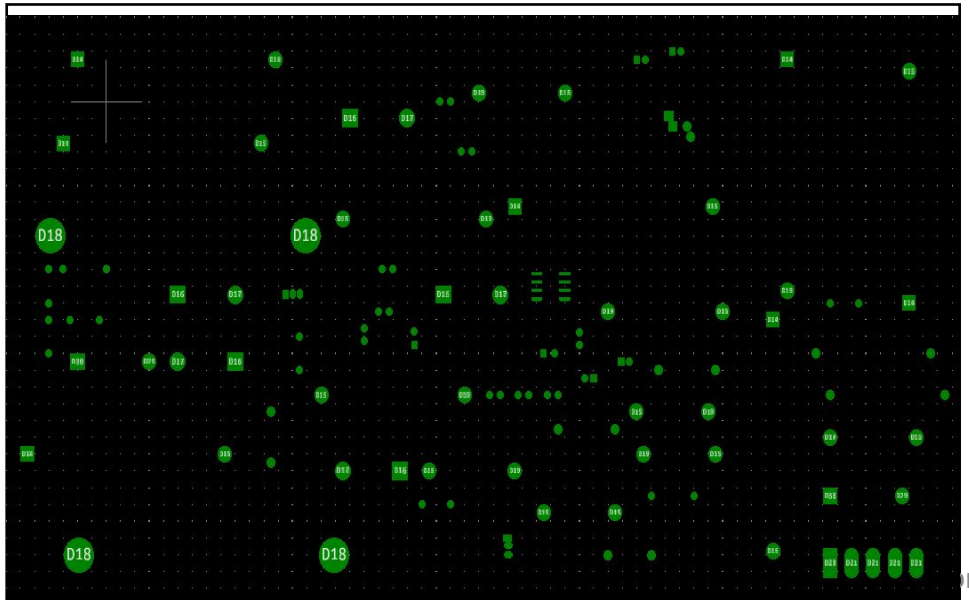
DRILL FILE



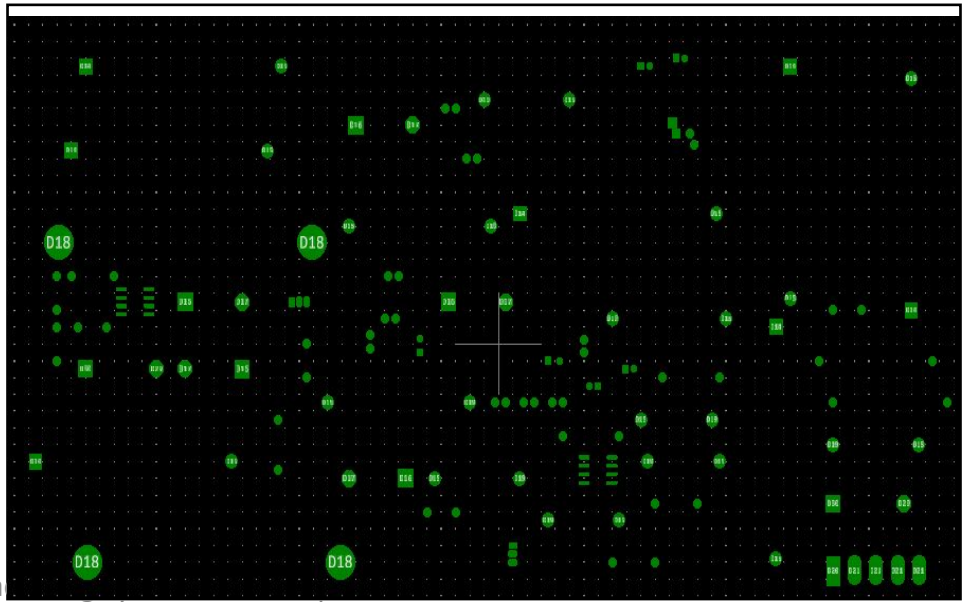
EDGE CUT



FRONT MASK

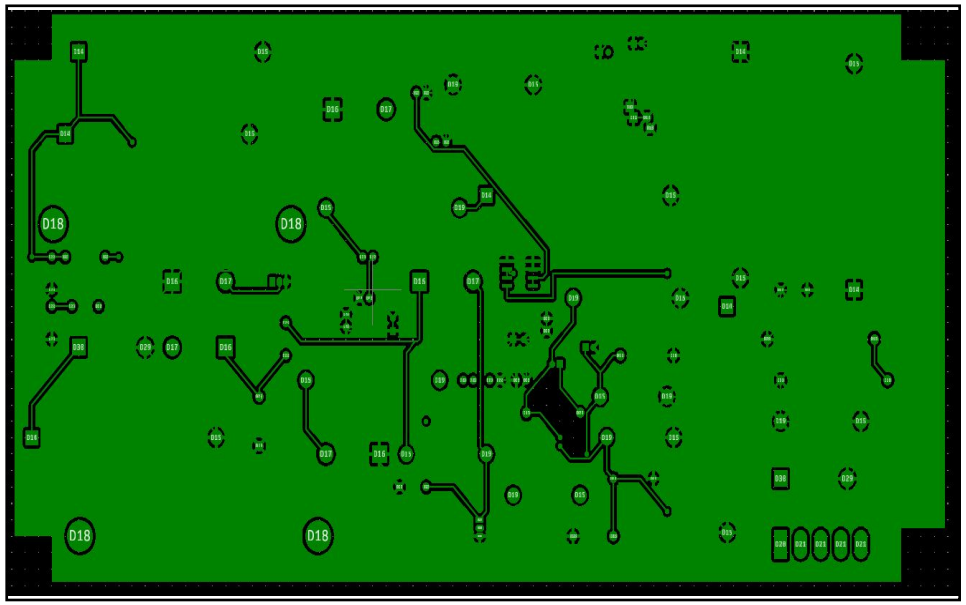


BACK MASK

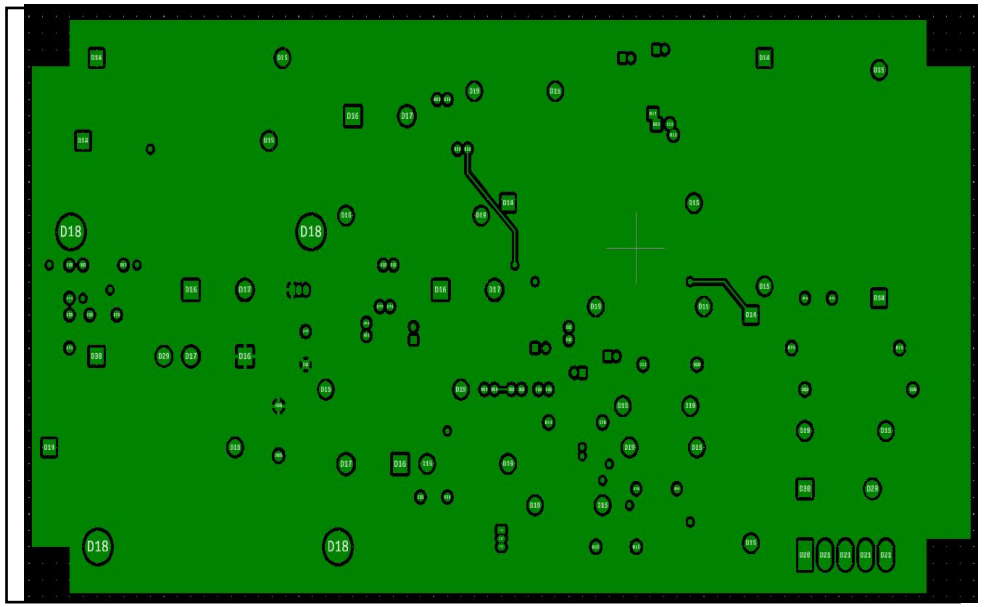


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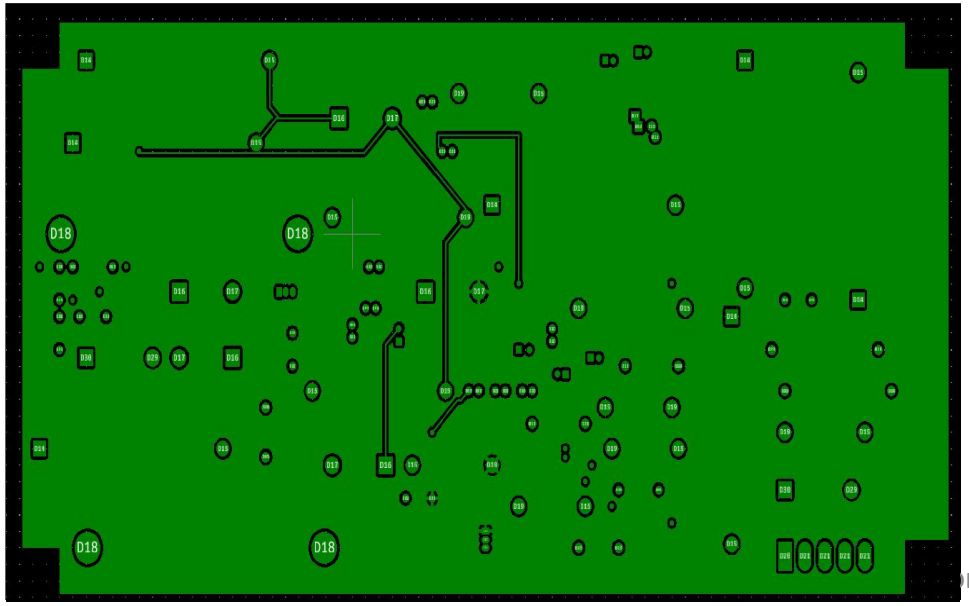
LAYER 1



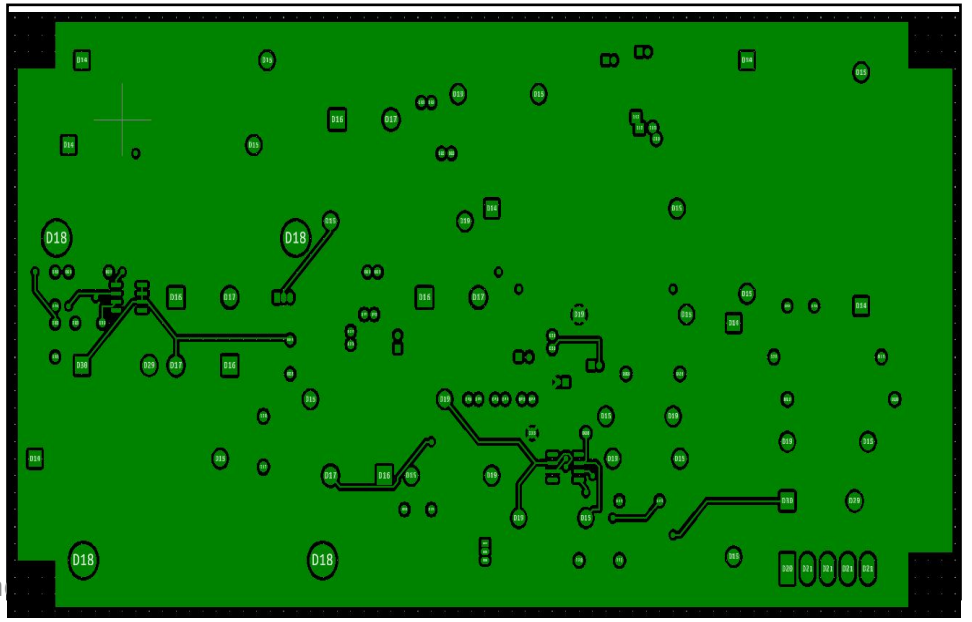
LAYER 2



LAYER 3

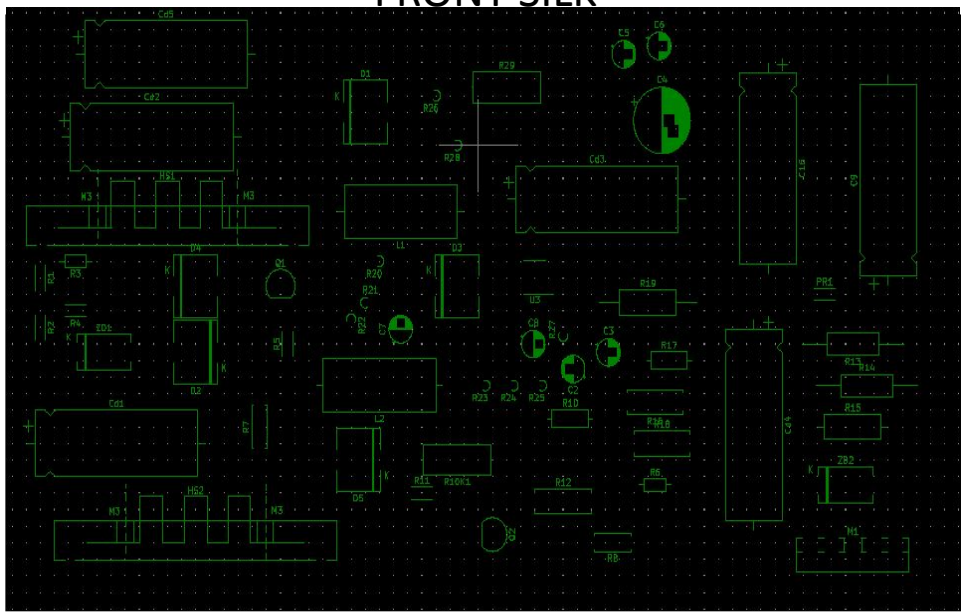


LAYER 4

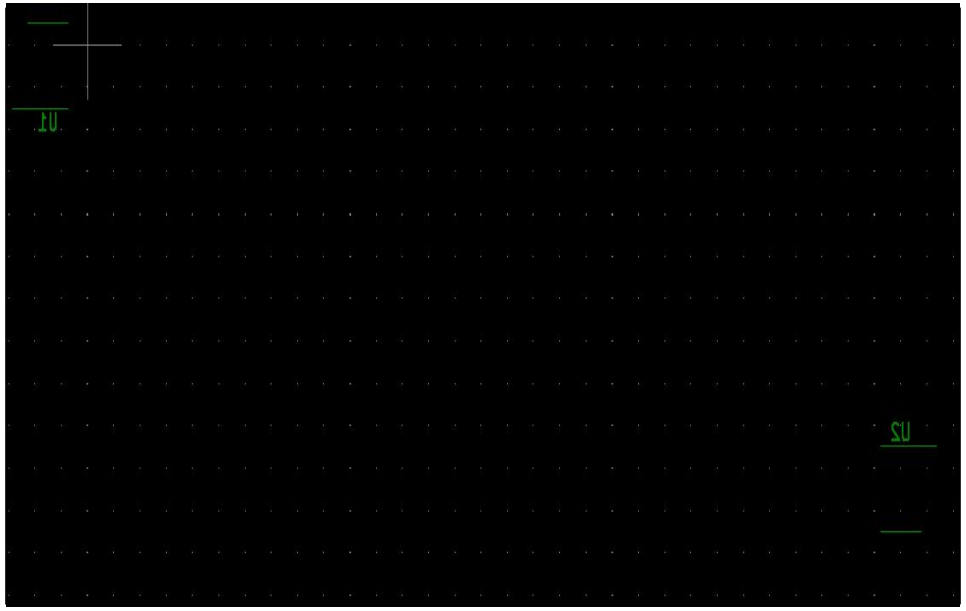


GERBER FILES

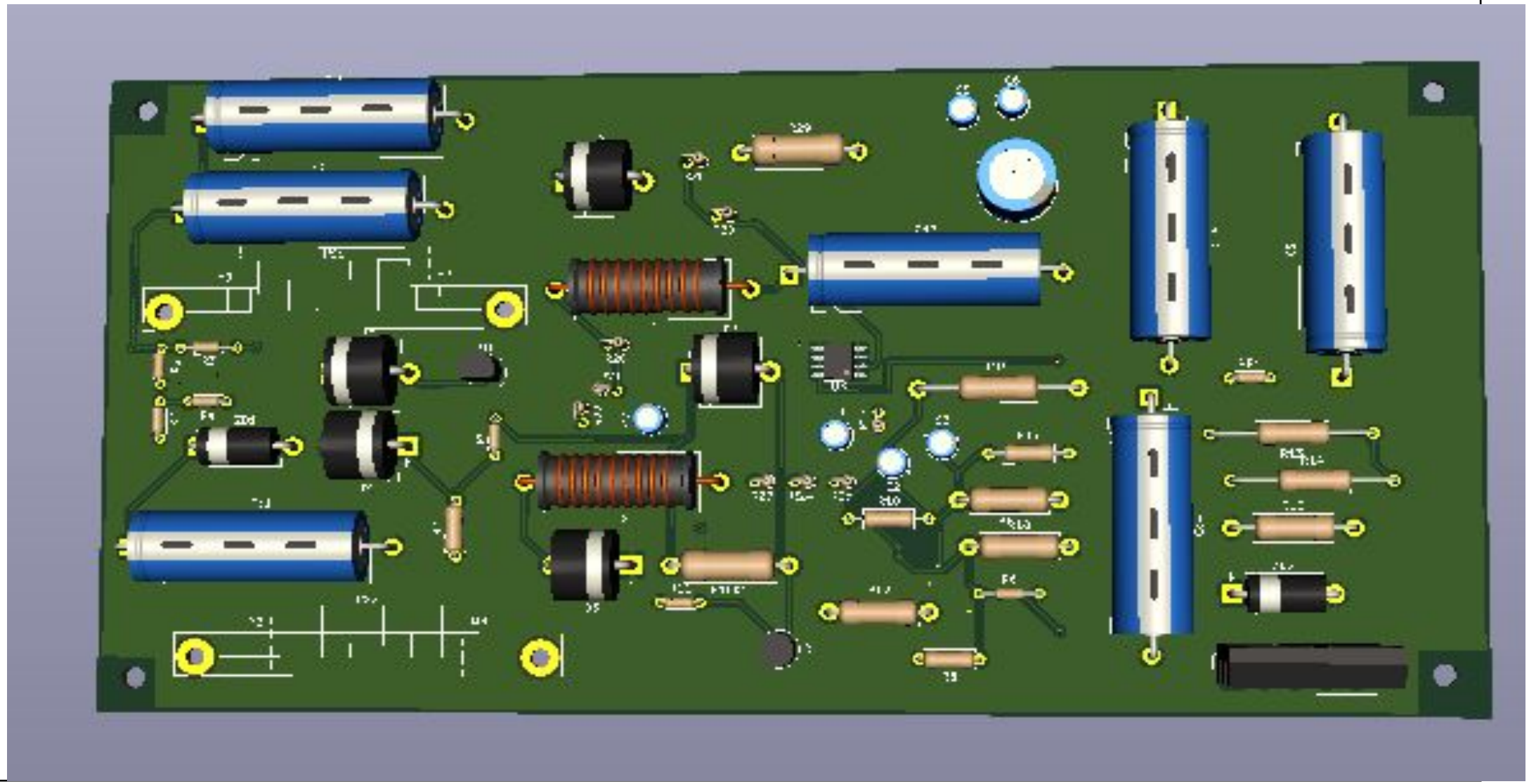
FRONT SILK



BACK SILK



3D VIEW



PCB DESIGN STEPS IN Ki-CAD

Step1: Draw a schematic .

Step2: Assign values to the components used.

Step3: Annotate the schematic.

Step4:Perform electrical rules check.

Step5:Assign footprints to each components.

Step6: Generate netlist.

Step7:Open PCB editor and import the netlist.

Step8:Give routing.

Step9:Generate drill and gerber files.

Any other Learning’s & take away’s